

Enterprise E-Commerce Business Management Application

Complete A-Z Knowledge Guide (Angular + Spring Boot)

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1. Application Overview

An Enterprise E-Commerce Business Management System (EBMS) is a centralized platform that manages every aspect of an online retail business — from product listing and inventory, to customer orders, payments, logistics, returns, and analytics. It serves internal teams (admin, operations, warehouse, finance, support) and integrates with external systems (payment gateways, shipping carriers, marketplaces, ERP).

Key Principles

- **Role-Based Access Control (RBAC):** Every feature is gated by role and permission.
 - **Audit Trail:** All critical actions are logged with user, timestamp, and before/after state.
 - **Event-Driven Flow:** Module transitions trigger downstream actions automatically.
 - **Multi-tenant Ready:** Supports multiple storefronts, warehouses, and currency zones.
 - **Scalable Architecture:** Microservice-friendly modular Spring Boot backend with Angular SPA frontend.
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2. Core Modules

2.1 Authentication & User Management Module

Handles login, registration, role assignment, password policies, session management, and MFA.

Sub-components:

- Login / Logout
 - MFA (Multi-Factor Authentication)
 - Password reset & expiry
 - User creation, editing, deactivation
 - Role and permission assignment
 - Session tracking and forced logout
 - Activity log per user
-

2.2 Dashboard & Analytics Module

The central home screen for all roles. Provides KPIs, charts, and quick-action shortcuts tailored per role.

Sub-components:

- Executive Dashboard (revenue, orders, customers, inventory)
- Operations Dashboard (pending orders, shipments, returns)

- Finance Dashboard (payables, receivables, refunds)
 - Warehouse Dashboard (stock alerts, pick queue)
 - Custom Widget Builder
 - Date range filters and comparison views
-

2.3 Catalog & Product Management Module

Manages all products, variants, pricing, media, and categorization.

Sub-components:

- Product creation and editing
 - Product variants (size, color, material, etc.)
 - SKU and barcode management
 - Categories and sub-categories
 - Brands and manufacturers
 - Pricing tiers (retail, wholesale, promo)
 - Product media (images, videos, 360° view)
 - SEO metadata per product
 - Product bundles and kits
 - Digital products (downloadable)
 - Product import/export (CSV, Excel)
 - Product approval workflow
-

2.4 Inventory Management Module

Tracks stock levels across all warehouses and locations in real time.

Sub-components:

- Stock-on-hand per SKU per location
- Stock reservations (items in open orders)

- Available-to-promise (ATP) calculation
 - Reorder point and reorder quantity configuration
 - Purchase order auto-suggestion
 - Stock transfer between warehouses
 - Stock adjustments (damage, shrinkage, found)
 - Cycle count and physical inventory
 - Batch/Lot and expiry date tracking
 - Inventory valuation (FIFO, LIFO, Weighted Average)
 - Low stock and out-of-stock alerts
-

2.5 Supplier & Purchase Order Module

Manages vendor relationships and the procurement lifecycle.

Sub-components:

- Supplier master data (contact, terms, rating)
 - Request for Quotation (RFQ)
 - Purchase Order (PO) creation and approval
 - PO line item tracking
 - Goods Receipt Note (GRN)
 - Supplier invoice matching (3-way match: PO → GRN → Invoice)
 - Supplier performance scorecard
 - Approved supplier list (ASL)
-

2.6 Customer Management Module (CRM-lite)

Manages customer profiles, segments, purchase history, and communication.

Sub-components:

- Customer profile (B2C and B2B)

- Customer groups and segments
 - Address book management
 - Order history per customer
 - Customer tier (Regular, Silver, Gold, Platinum)
 - Wishlist and saved carts
 - Customer communication log
 - Customer credit limit (B2B)
 - Blacklist and fraud flags
-

2.7 Order Management Module (OMS)

The core transactional module — manages the full lifecycle of a customer order.

Sub-components:

- Order creation (storefront, manual, API, import)
 - Order detail view (items, pricing, taxes, discounts)
 - Order editing (pre-fulfillment)
 - Order splitting (partial fulfillment)
 - Order merging
 - Order hold and release
 - Fraud check and order risk scoring
 - Tax calculation per jurisdiction
 - Discount and coupon application
 - Order timeline/audit log
-

2.8 Payment Management Module

Processes and tracks all payment transactions, refunds, and reconciliation.

Sub-components:

- Payment gateway integration (Stripe, PayPal, Razorpay, etc.)
 - Payment method management (card, UPI, wallet, COD, net banking)
 - Payment capture and authorization
 - Partial payments and installments
 - Refund processing (full and partial)
 - Payment reconciliation report
 - Chargeback and dispute management
 - Payout management (marketplace sellers)
 - Payment link generation
-

2.9 Fulfillment & Warehouse Management Module (WMS)

Manages the pick, pack, and ship process within the warehouse.

Sub-components:

- Fulfillment queue and priority management
 - Pick list generation (single, batch, zone, wave)
 - Packing station and carton management
 - Label printing (shipping, packing slips, barcode)
 - Quality check (QC) at packing
 - Bin and shelf location mapping
 - Returns processing (receive, inspect, restock/dispose)
 - Warehouse worker task assignment
-

2.10 Shipping & Logistics Module

Manages carrier integrations, shipment tracking, and delivery SLAs.

Sub-components:

- Shipping carrier integration (FedEx, UPS, DHL, DTDC, etc.)

- Shipping rate shopping and comparison
 - Shipment creation and tracking number assignment
 - Real-time tracking updates (webhook/polling)
 - Delivery SLA monitoring
 - Proof of delivery (POD)
 - Failed delivery and re-attempt management
 - Shipping zone and rate configuration
 - Freight cost allocation
-

2.11 Returns & Refund Management Module (RMA)

Manages the full post-sale reverse logistics cycle.

Sub-components:

- Return Merchandise Authorization (RMA) creation
 - Return reason capture
 - Return pickup scheduling
 - Return receipt and inspection
 - Restocking or disposal decision
 - Refund or replacement initiation
 - Store credit issuance
 - Return rate reporting
-

2.12 Promotions & Marketing Module

Manages discounts, coupons, loyalty programs, and campaign tracking.

Sub-components:

- Coupon code creation and rules
- Automatic discounts (cart rules, volume tiers)

- Flash sales and scheduled pricing
 - Loyalty/rewards points program
 - Gift cards and store credits
 - Referral programs
 - Email and SMS campaign management
 - Abandoned cart recovery
 - Banner and promotional content management
-

2.13 Finance & Accounting Module

Tracks revenue, expenses, payables, receivables, and financial reports.

Sub-components:

- Sales revenue tracking
 - Cost of Goods Sold (COGS) tracking
 - Accounts payable (supplier invoices)
 - Accounts receivable (customer payments)
 - Tax management (GST, VAT, Sales Tax)
 - Financial reconciliation
 - Profit and loss (P&L) view
 - Expense management
 - Currency and exchange rate management
 - Export to accounting systems (Tally, QuickBooks, SAP)
-

2.14 Reporting & Analytics Module

Centralized reports and BI dashboards across all modules.

Sub-components:

- Sales reports (by date, product, category, channel)

- Inventory reports (stock value, movement, ageing)
 - Order reports (fulfillment rate, cancellation rate)
 - Customer reports (LTV, cohort, churn)
 - Financial reports (revenue, margin, tax summary)
 - Shipping reports (on-time delivery, carrier performance)
 - Return reports (return rate, reason analysis)
 - Custom report builder
 - Scheduled report delivery (email/PDF)
 - Export to Excel / CSV
-

2.15 Notification & Communication Module

Manages all system-generated and manual communications.

Sub-components:

- Email notification templates
 - SMS notification templates
 - Push notification management
 - In-app alerts and bell notifications
 - Notification event triggers configuration
 - Communication logs
 - WhatsApp Business integration
-

2.16 System Configuration & Settings Module

Admin-only module for configuring platform-wide settings.

Sub-components:

- Store settings (name, logo, timezone, currency, language)
- Tax configuration (rates, rules, jurisdictions)

- Shipping zone and rate setup
 - Payment gateway configuration
 - Integration management (API keys, webhooks)
 - Email and SMTP settings
 - Audit log viewer
 - Feature flags and toggle management
 - Backup and restore settings
-

3. Module Inter-Flow & Dependencies

```
[Customer Places Order]
    |
    v
[Order Management Module]
  - Validates stock availability → Inventory Module
  - Applies promotions/coupons → Promotions Module
  - Calculates tax → Finance Module
  - Runs fraud check → Payment Module
    |
    v
[Payment Management Module]
  - Authorizes or captures payment
  - On success → triggers fulfillment
  - On failure → order stays in PENDING_PAYMENT
    |
    v
[Fulfillment & WMS Module]
  - Generates pick list → Warehouse workers
  - Items picked and packed
  - Deducts stock → Inventory Module
  - Label printed → Shipping Module
    |
    v
[Shipping & Logistics Module]
  - Shipment created with carrier
  - Tracking number assigned to order
  - Real-time tracking events updated
    |
    v
```

```

[Order Delivered]
- Order status → DELIVERED
- Customer notification sent → Notification Module
- Payment settled → Finance Module
  |
  v
[Returns (if any)]
- Customer initiates return → RMA Module
- Return received → WMS Module
- Inspection done → Inventory Module (restock/dispose)
- Refund processed → Payment Module
- Finance updated → Finance Module

```

Module Dependency Matrix

Module	Depends On
Order Management	Inventory, Payment, Promotions, Customer, Catalog
Fulfillment/WMS	Order, Inventory, Shipping
Shipping	Fulfillment, Order
Payment	Order, Finance
Returns/RMA	Order, Payment, Inventory, Shipping
Finance	Order, Payment, Inventory, Supplier/PO
Reporting	ALL modules
Notifications	ALL modules (event-driven)
Inventory	Catalog, Supplier/PO, WMS

4. Security Architecture

4.1 Authentication

- **JWT (JSON Web Token):** Stateless authentication. Access token (15–30 min TTL) + Refresh token (7–30 days TTL).
- **Multi-Factor Authentication (MFA):** TOTP (Google Authenticator) or OTP via

SMS/email.

- **OAuth2 / SSO:** Integration with Google Workspace, Microsoft Azure AD for enterprise SSO.
- **Spring Security:** Configures security filter chains, CORS, CSRF, and request authorization.

4.2 Authorization

- **RBAC (Role-Based Access Control):** Roles define coarse-grained access.
- **Permission-Based (Fine-Grained):** Each feature/action has an explicit permission flag.
- **Spring Method Security:** `@PreAuthorize`, `@PostAuthorize` annotations on service methods.
- **Angular Route Guards:** `CanActivate` guards check role/permission before loading pages.

4.3 Data Security

- **Field-Level Encryption:** Sensitive fields (card data, SSN, passwords) are encrypted at rest.
- **BCrypt Password Hashing:** All passwords hashed with BCrypt (strength 12+).
- **PII Data Masking:** Customer PII masked in logs and non-prod environments.
- **HTTPS Everywhere:** TLS 1.2+ enforced on all endpoints.
- **Data Isolation:** Multi-tenant data separated by tenant ID at DB query level.

4.4 API Security

- **Rate Limiting:** Per-user and per-IP request throttling (e.g., 100 req/min).
- **Input Validation:** All inputs validated server-side (Bean Validation, Hibernate Validator).
- **SQL Injection Prevention:** JPA/Hibernate parameterized queries. No raw SQL from user input.
- **XSS Prevention:** Content Security Policy (CSP) headers. Angular built-in XSS sanitization.
- **CSRF Protection:** CSRF tokens for session-based endpoints (or disabled for stateless JWT APIs).

- **Sensitive Endpoint Hardening:** Admin endpoints IP-whitelisted.

4.5 Audit & Compliance

- **Audit Log:** Every create/update/delete recorded with: user, timestamp, IP, old value, new value.
- **GDPR Compliance:** Right to erasure, data export, consent management.
- **PCI-DSS:** Card data never stored locally; tokenized via payment gateway.
- **Session Management:** Concurrent session limits, forced logout on password change.

4.6 Infrastructure Security

- **API Gateway:** All traffic routed through a gateway (Kong/AWS API Gateway) for auth, rate-limit, logging.
 - **Secrets Management:** Vault (HashiCorp) or AWS Secrets Manager. No secrets in code/config files.
 - **Container Security:** Docker images scanned. Non-root user in containers.
 - **Network Segmentation:** DB accessible only from app server. No direct public DB access.
-

5. User Roles & Activities

5.1 Super Admin

Description: Full platform access. Sets up the application, manages tenants, and configures system-wide settings.

Activities:

- Create and manage all user accounts and roles
- Configure system settings and feature flags
- Manage all integrations and API keys
- View and export all audit logs
- Access all modules without restriction
- Manage backup and restore

- Impersonate other users for support (logged)
-

5.2 Admin

Description: Business administrator. Manages the day-to-day operations and staff.

Activities:

- Create and manage staff user accounts
 - Assign roles and permissions to staff
 - Configure store settings, tax, and shipping zones
 - View all dashboards and reports
 - Approve high-value orders and POs
 - Manage promotions and discount rules
 - Override order statuses and payments
-

5.3 Product Manager

Description: Manages the product catalog, pricing, and inventory planning.

Activities:

- Create, edit, and publish products
 - Manage product categories, brands, and attributes
 - Set and modify pricing and promotions
 - Configure product variants and bundles
 - Import/export product data
 - Monitor inventory levels and set reorder points
 - Initiate purchase orders for low-stock items
-

5.4 Inventory Manager

Description: Controls stock levels, adjustments, and warehouse transfers.

Activities:

- View and adjust stock levels
 - Perform cycle counts and physical inventory
 - Initiate and approve stock transfers
 - Configure reorder rules
 - View inventory reports and stock ageing
 - Manage batch/lot and expiry tracking
 - Receive and process goods receipt notes (GRN)
-

5.5 Purchase Manager

Description: Handles procurement from supplier selection through PO and GRN.

Activities:

- Create and manage supplier master data
 - Create, edit, and submit RFQs
 - Create and approve Purchase Orders
 - Process Goods Receipt Notes
 - Perform 3-way invoice matching
 - View supplier performance reports
 - Manage approved supplier list
-

5.6 Order Manager / Operations Staff

Description: Manages order lifecycle from placement to fulfillment.

Activities:

- View and manage all orders
- Edit orders (pre-fulfillment)
- Put orders on hold or release them

- Cancel and split orders
 - Manually create orders for customers
 - Assign orders to fulfillment queues
 - Escalate high-risk or disputed orders
-

5.7 Warehouse Staff / Picker-Packer

Description: Executes physical warehouse tasks — picking, packing, labeling.

Activities:

- View assigned pick lists
 - Confirm item picks (barcode scan or manual)
 - Pack orders and record carton details
 - Print shipping labels and packing slips
 - Perform quality checks
 - Record discrepancies during picking
 - Process inbound returns
-

5.8 Shipping Coordinator

Description: Manages carrier selection, shipment creation, and delivery tracking.

Activities:

- Create shipments and assign carriers
- Compare shipping rates
- Monitor delivery SLAs
- Manage failed deliveries and re-attempts
- Update tracking information manually
- Liaise with carriers for disputes
- View shipping performance reports

5.9 Finance / Accounts Staff

Description: Manages payments, reconciliation, refunds, and financial reporting.

Activities:

- View and process payment transactions
 - Initiate and approve refunds
 - Handle chargebacks and disputes
 - Reconcile payments with orders
 - Manage supplier invoice payments
 - Generate tax reports
 - Export financial data to accounting systems
-

5.10 Customer Support Representative

Description: Handles customer queries, order issues, returns, and escalations.

Activities:

- Search and view customer profiles and order history
 - Create RMAs (returns/refunds) on behalf of customers
 - Apply store credits or issue gift cards
 - Put orders on hold or cancel (with reason)
 - Update customer address and contact details
 - Log communication notes on customer records
 - View customer communication history
-

5.11 Marketing Manager

Description: Creates and manages promotions, campaigns, and customer engagement.

Activities:

- Create and manage coupon codes and discount rules

- Schedule flash sales and time-limited offers
 - Manage loyalty program rules and points
 - Design and send email/SMS campaigns
 - Manage banner and promotional content
 - View campaign performance analytics
 - Manage abandoned cart recovery rules
-

5.12 Read-Only Analyst / Auditor

Description: View-only access to reports and data for auditing or analysis.

Activities:

- View all dashboards and reports
 - Export data to CSV/Excel
 - View audit logs
 - No create, edit, or delete capabilities
-

6. Module Statuses & Lifecycle

6.1 Order Statuses

Status	Description
DRAFT	Order being created but not yet submitted
PENDING_PAYMENT	Order placed; awaiting payment authorization
PAYMENT_FAILED	Payment attempt was unsuccessful
CONFIRMED	Payment captured; order confirmed
ON_HOLD	Order paused (fraud review, stock issue, manual hold)
PROCESSING	Order being prepared for fulfillment

PARTIALLY_FULFILLED	Some items shipped; others pending
FULFILLED	All items picked and packed; ready for dispatch
SHIPPED	Handed over to carrier; tracking active
OUT_FOR_DELIVERY	Last-mile delivery in progress
DELIVERED	Customer has received the order
CANCELLED	Order cancelled (before shipment)
RETURN_INITIATED	Customer has requested a return
RETURN_IN_TRANSIT	Return shipment on its way back
RETURN_RECEIVED	Return arrived at warehouse
REFUNDED	Refund processed to customer
PARTIALLY_REFUNDED	Partial refund processed
CLOSED	Order fully settled; no further action

6.2 Payment Statuses

Status	Description
PENDING	Payment not yet attempted
AUTHORIZED	Payment authorized but not captured
CAPTURED	Payment successfully captured
PARTIALLY_CAPTURED	Partial amount captured
FAILED	Payment attempt failed
VOIDED	Authorization voided before capture
REFUND_PENDING	Refund initiated; awaiting processing
REFUNDED	Refund successfully processed
PARTIALLY_REFUNDED	Partial refund completed

CHARGEBACK	Customer raised a dispute with bank
DISPUTED	Dispute under review

6.3 Inventory / Stock Statuses

Status	Description
IN_STOCK	Available quantity > 0
LOW_STOCK	Quantity below reorder point threshold
OUT_OF_STOCK	Available quantity = 0
RESERVED	Quantity committed to open orders
IN_TRANSIT	Stock en route from supplier or warehouse
ON_HOLD	Stock quarantined (QC, damage)
DISCONTINUED	Product no longer restocked

6.4 Product Statuses

Status	Description
DRAFT	Being created; not visible in storefront
PENDING_APPROVAL	Submitted for review
APPROVED	Reviewed and approved
ACTIVE	Live and visible on storefront
INACTIVE	Temporarily hidden from storefront
ARCHIVED	Discontinued and removed from listing
REJECTED	Failed approval review

6.5 Purchase Order (PO) Statuses

Status	Description
DRAFT	Being created; not sent to supplier
PENDING_APPROVAL	Awaiting manager approval
APPROVED	Approved and sent to supplier
ACKNOWLEDGED	Supplier has confirmed the PO
PARTIALLY_RECEIVED	Some goods received
FULLY_RECEIVED	All goods received
INVOICED	Supplier invoice matched to PO
PAID	Payment made to supplier
CANCELLED	PO cancelled
CLOSED	All items received and settled

6.6 Return / RMA Statuses

Status	Description
REQUESTED	Customer has requested a return
APPROVED	Return approved; instructions sent
REJECTED	Return not approved
PICKUP_SCHEDULED	Return pickup arranged
IN_TRANSIT	Return being shipped back
RECEIVED	Return received at warehouse
UNDER_INSPECTION	Quality check in progress
RESTOCKED	Item returned to inventory
DISPOSED	Item damaged; written off

REFUND_INITIATED	Refund triggered after inspection
COMPLETED	Full return lifecycle done

6.7 Shipment Statuses

Status	Description
LABEL_CREATED	Shipment label generated
PICKED_UP	Carrier has collected the parcel
IN_TRANSIT	Moving through carrier network
OUT_FOR_DELIVERY	Last mile; with delivery agent
DELIVERED	Successfully delivered
ATTEMPTED_DELIVERY	Delivery tried; no one available
RETURNED_TO_SENDER	Carrier returned the parcel
LOST	Shipment lost; investigation open
EXCEPTION	Unexpected delay or issue

6.8 Supplier Statuses

Status	Description
PROSPECT	Under evaluation
ACTIVE	Approved and actively used
ON_HOLD	Temporarily paused
BLACKLISTED	Blocked from new POs
INACTIVE	No longer engaged

6.9 User Account Statuses

Status	Description
ACTIVE	Can log in and operate
INACTIVE	Manually deactivated
LOCKED	Too many failed login attempts
PENDING_VERIFICATION	Email/phone not yet verified
SUSPENDED	Suspended by admin due to violation

7. Sidebar Menu Structure

Each menu item lists which roles can see it and what actions it enables.



Dashboard

- **Overview Dashboard** — All roles (role-specific widgets)
- **Executive Summary** — Super Admin, Admin
- **Operations Board** — Order Manager, Warehouse Staff
- **Finance Summary** — Finance Staff, Admin



Catalog

- **Products**
 - All Products (list, search, filter)
 - Add New Product
 - Bulk Import Products
 - Archived Products
- **Categories**
 - Category Tree

- Add / Edit Category
- **Brands**
- **Attributes & Variants**
- **Product Bundles**
- **Digital Products**
- **Product Approvals** (Admin / Product Manager)

Roles: Super Admin, Admin, Product Manager



Inventory

- **Stock Overview** (by warehouse, by SKU)
- **Stock Adjustments**
- **Stock Transfers**
- **Reorder Rules**
- **Low Stock Alerts**
- **Batch / Lot Tracking**
- **Cycle Count**
- **Inventory Valuation Report**

Roles: Super Admin, Admin, Inventory Manager, Product Manager (view)



Suppliers

- **Supplier Directory**
- **Add New Supplier**
- **Request for Quotation (RFQ)**
- **Purchase Orders**
 - All POs
 - Create PO

- PO Approval Queue
- **Goods Receipt Notes (GRN)**
- **Supplier Invoices**
- **Supplier Performance**

Roles: Super Admin, Admin, Purchase Manager, Inventory Manager (GRN only)

Customers

- **Customer Directory**
- **Add Customer**
- **Customer Groups & Segments**
- **Customer Communication Log**
- **Loyalty & Rewards**
- **Blacklisted Customers**
- **B2B Accounts & Credit Limits**

Roles: Super Admin, Admin, Customer Support, Marketing Manager

Orders

- **All Orders**
- **Create Manual Order**
- **Order Approval Queue** (high-value / flagged)
- **On Hold Orders**
- **Cancelled Orders**
- **Order Import**

Roles: Super Admin, Admin, Order Manager, Customer Support

Payments

- **Transaction List**
- **Pending Payments**
- **Refunds**
 - Refund Requests
 - Process Refund
 - Refund History
- **Chargebacks & Disputes**
- **Payment Reconciliation**
- **Payouts** (marketplace sellers)
- **Payment Links**

Roles: Super Admin, Admin, Finance Staff



Warehouse (WMS)

- **Fulfillment Queue**
- **Pick Lists**
 - Generate Pick List
 - My Pick Tasks
- **Packing Station**
- **Label Printing**
- **Quality Check**
- **Bin & Location Management**
- **Inbound Returns Processing**

Roles: Super Admin, Admin, Order Manager, Warehouse Staff



Shipping

- **Shipments**

- All Shipments
- Create Shipment
- Bulk Ship
- **Tracking Monitor**
- **Failed Deliveries**
- **Carrier Management**
- **Shipping Rates & Zones**
- **Proof of Delivery (POD)**
- **Shipping Reports**

Roles: Super Admin, Admin, Shipping Coordinator, Order Manager



Returns (RMA)

- **Return Requests**
- **Approve / Reject Returns**
- **Returns In Transit**
- **Return Inspection**
- **Refund / Replacement**
- **Return Analytics**

Roles: Super Admin, Admin, Customer Support, Warehouse Staff, Finance Staff



Promotions & Marketing

- **Coupons**
 - All Coupons
 - Create Coupon
- **Automatic Discounts**
- **Flash Sales**

- **Loyalty Program**
- **Gift Cards & Store Credits**
- **Referral Programs**
- **Campaigns**
 - Email Campaigns
 - SMS Campaigns
- **Abandoned Cart Recovery**
- **Banners & Content**

Roles: Super Admin, Admin, Marketing Manager

Finance

- **Revenue Overview**
- **Accounts Payable**
- **Accounts Receivable**
- **Tax Management**
 - Tax Rates
 - Tax Reports
- **Expense Tracking**
- **P&L Statement**
- **Financial Reconciliation**
- **Export to Accounting System**

Roles: Super Admin, Admin, Finance Staff

Reports & Analytics

- **Sales Reports**
- **Inventory Reports**

- **Order Reports**
- **Customer Reports**
- **Finance Reports**
- **Shipping Reports**
- **Return Reports**
- **Custom Report Builder**
- **Scheduled Reports**

Roles: Super Admin, Admin, Analyst/Auditor, and relevant module managers (scoped)

Notifications

- **Notification Templates**
- **Notification Events Config**
- **Communication Logs**
- **Push Notification Manager**

Roles: Super Admin, Admin

Settings

- **Store Settings**
- **User Management**
 - All Users
 - Add User
 - Roles & Permissions
- **Tax Configuration**
- **Shipping Configuration**
- **Payment Gateway Settings**
- **Integration Manager**

- API Keys
- Webhooks
- Marketplace Connections
- **Email / SMTP Settings**
- **Feature Flags**
- **Audit Logs**
- **Backup & Restore**

Roles: Super Admin, Admin (scoped)

8. Angular Pages & Activities

8.1 Auth Pages

Page	Route	Activity
Login	/auth/login	Enter credentials, MFA verification, redirect on success
Forgot Password	/auth/forgot-password	Enter email, receive reset link
Reset Password	/auth/reset-password	Enter new password via token link
MFA Setup	/auth/mfa-setup	Scan QR code, enter OTP, confirm setup
Session Expired	/auth/session-expired	Inform user; redirect to login

8.2 Dashboard Pages

Page	Route	Activity
Main Dashboard	/dashboard	View KPIs, charts, alerts, quick actions

Executive Dashboard	/dashboard/executive	Revenue trend, top products, customer growth
Operations Dashboard	/dashboard/operations	Order queue, SLA alerts, fulfillment status

8.3 Catalog Pages

Page	Route	Activity
Product List	/catalog/products	Search, filter, sort, bulk actions, export
Add Product	/catalog/products/new	Multi-step form: details, variants, pricing, media, SEO
Edit Product	/catalog/products/:id/edit	Edit all product fields; submit for approval
Product Detail	/catalog/products/:id	Read-only view with full detail
Category List	/catalog/categories	Tree view, drag-and-drop ordering
Add/Edit Category	/catalog/categories/new , /catalog/categories/:id/edit	Category form with parent, image, SEO
Brand List	/catalog/brands	List, add, edit, deactivate brands
Attributes	/catalog/attributes	Create attribute groups and values
Bundles	/catalog/bundles	Create product kits and bundles
Product Import	/catalog/products/import	Upload CSV/Excel, map columns, preview, confirm
Approval Queue	/catalog/approvals	List pending products; approve or reject with notes

8.4 Inventory Pages

Page	Route	Activity
Stock Overview	<code>/inventory/stock</code>	View stock by SKU, warehouse, filter by status
Stock Detail	<code>/inventory/stock/:sku</code>	Full stock history and movement for one SKU
Stock Adjustment	<code>/inventory/adjustments/new</code>	Select SKU, quantity, reason, submit
Adjustment History	<code>/inventory/adjustments</code>	Filter by date, user, reason
Stock Transfer	<code>/inventory/transfers/new</code>	Select source/destination warehouse, SKUs, qty
Transfer History	<code>/inventory/transfers</code>	View all transfers and statuses
Reorder Rules	<code>/inventory/reorder-rules</code>	Set reorder point and quantity per SKU
Cycle Count	<code>/inventory/cycle-count</code>	Select location, list expected vs actual, submit variances
Low Stock Alerts	<code>/inventory/alerts</code>	Filtered list of SKUs below reorder point

8.5 Supplier & Purchase Pages

Page	Route	Activity
Supplier List	<code>/suppliers</code>	Search, filter, view status
Add/Edit Supplier	<code>/suppliers/new ,</code> <code>/suppliers/:id/edit</code>	Full supplier form with contacts, payment terms
Supplier Detail	<code>/suppliers/:id</code>	View PO history, performance, contacts
RFQ List	<code>/suppliers/rfq</code>	List all RFQs by status
		Select supplier, items, quantities, due

Create RFQ	/suppliers/rfq/new	date
PO List	/purchase-orders	Filter by status, supplier, date
Create PO	/purchase-orders/new	Select supplier, add line items, set terms
PO Detail	/purchase-orders/:id	View PO, approve/reject, print PO
GRN List	/purchase-orders/grn	All goods receipts
Create GRN	/purchase-orders/:poId/grn/new	Record received quantities, damage notes
Invoice Matching	/purchase-orders/:id/invoice	3-way match PO, GRN, and supplier invoice

8.6 Customer Pages

Page	Route	Activity
Customer List	/customers	Search, filter by tier, segment, status
Add Customer	/customers/new	Create B2C or B2B customer profile
Customer Detail	/customers/:id	Profile, orders, addresses, communication log
Edit Customer	/customers/:id/edit	Update profile, address, credit limit
Customer Groups	/customers/groups	Create and manage segments
Communication Log	/customers/:id/communications	View all emails, SMS, notes for customer
Blacklist	/customers/blacklist	View and manage flagged/blocked customers

8.7 Order Pages

Page	Route	Activity
Order List	<code>/orders</code>	Search by ID, customer, status, date; bulk actions
Create Order	<code>/orders/new</code>	Select customer, add products, apply discount, select payment/shipping
Order Detail	<code>/orders/:id</code>	Full order view: items, status timeline, payments, shipments
Edit Order	<code>/orders/:id/edit</code>	Modify items, shipping address, discounts (pre-fulfillment only)
Order Hold/Release	(Action on detail page)	Add hold reason; release when resolved
Order Cancel	(Action on detail page)	Cancel with reason; triggers refund flow
Order Split	<code>/orders/:id/split</code>	Divide order into multiple fulfillments
Order Approval Queue	<code>/orders/approvals</code>	Review flagged or high-value orders

8.8 Payment Pages

Page	Route	Activity
Transaction List	<code>/payments/transactions</code>	Filter by status, method, date range
Transaction Detail	<code>/payments/transactions/:id</code>	Full transaction detail, related order
Refund List	<code>/payments/refunds</code>	Filter pending, processed, failed
Create Refund	<code>/payments/refunds/new</code>	Select order, amount (full/partial), reason
Chargeback List	<code>/payments/chargebacks</code>	View disputes, respond to disputes

Reconciliation	/payments/reconciliation	Match gateway settlements with internal records
Payment Links	/payments/payment-links	Generate and share payment links

8.9 Warehouse (WMS) Pages

Page	Route	Activity
Fulfillment Queue	/warehouse/queue	View orders ready to pick, prioritize
Pick Lists	/warehouse/pick-lists	List all generated pick lists
My Tasks	/warehouse/my-tasks	Worker's assigned picks
Pick Detail	/warehouse/pick-lists/:id	Item-by-item picking confirmation
Pack Station	/warehouse/pack/:orderId	Pack items, record carton, print label
Label Print	/warehouse/labels	Batch print shipping labels
QC Screen	/warehouse/qc/:orderId	Quality check items before dispatch
Returns Receiving	/warehouse/returns/receive	Scan return shipment, log condition
Bin Management	/warehouse/bins	Map and manage bin/shelf locations

8.10 Shipping Pages

Page	Route	Activity
Shipment List	/shipping/shipments	All shipments, filter by carrier/status/date
Create Shipment	/shipping/shipments/new	Select order, compare rates, assign carrier
Shipment Detail	/shipping/shipments/:id	Tracking history, carrier info, POD

Tracking Monitor	/shipping/tracking	Live tracking status board
Failed Deliveries	/shipping/failed	List of undelivered parcels; reschedule
Carrier Management	/shipping/carriers	Configure carriers, API credentials
Rate & Zone Config	/shipping/rates	Define zones and rates per carrier

8.11 Returns (RMA) Pages

Page	Route	Activity
Return Requests	/returns	All RMAs, filter by status
RMA Detail	/returns/:id	Full return timeline and actions
Approve/Reject	(Action on RMA detail)	Approve with pickup scheduling or reject with reason
Inspection	/returns/:id/inspect	Log condition, decide restock or dispose
Refund/Replace	(Action on RMA detail)	Trigger refund or create replacement order
Return Analytics	/returns/analytics	Return rate, top reasons, cost impact

8.12 Promotions & Marketing Pages

Page	Route	Activity
Coupon List	/marketing/coupons	View, search, enable/disable coupons
Create Coupon	/marketing/coupons/new	Define code, type, value, conditions, dates
Discount Rules	/marketing/discounts	Cart-level automatic discount rules

Flash Sales	/marketing/flash-sales	Schedule time-limited price drops
Loyalty Program	/marketing/loyalty	Points rules, redemption thresholds, member tiers
Gift Cards	/marketing/gift-cards	Issue, view, disable gift cards
Campaigns	/marketing/campaigns	Create email/SMS campaigns; view performance
Abandoned Cart	/marketing/abandoned-cart	View and configure recovery sequences
Banners	/marketing/banners	Create and schedule storefront banners

8.13 Finance Pages

Page	Route	Activity
Revenue Overview	/finance/revenue	Revenue by channel, product, period
Accounts Payable	/finance/payable	Supplier invoices, due dates, payment status
Accounts Receivable	/finance/receivable	Outstanding customer balances
Tax Management	/finance/tax	Configure rates, view tax collected
Tax Reports	/finance/tax/reports	GST/VAT summaries for filing
P&L Statement	/finance/pnl	Revenue vs COGS vs expenses
Reconciliation	/finance/reconciliation	Match all financial records
Export	/finance/export	Export to Tally, QuickBooks, SAP, Excel

8.14 Reports Pages

Page	Route	Activity

Report Hub	/reports	Browse all report categories
Sales Report	/reports/sales	Date/product/category/channel breakdown
Inventory Report	/reports/inventory	Stock value, movement, ageing
Order Report	/reports/orders	Fulfillment rate, cancellation trend
Customer Report	/reports/customers	LTV, cohort, retention, churn
Shipping Report	/reports/shipping	On-time %, carrier performance
Return Report	/reports/returns	Return rate, reason analysis
Custom Builder	/reports/builder	Drag-and-drop column/filter builder
Scheduled Reports	/reports/scheduled	Configure email delivery of reports

8.15 Settings Pages

Page	Route	Activity
Store Settings	/settings/store	Name, logo, timezone, currency, language
User Management	/settings/users	List all users, status, last login
Add/Edit User	/settings/users/new , /settings/users/:id/edit	Create user, assign role
Roles & Permissions	/settings/roles	Define roles, assign granular permissions
Tax Config	/settings/tax	Rates, jurisdictions, exemptions
Shipping Config	/settings/shipping	Zones, carrier config, default rules
Payment Config	/settings/payments	Gateway keys, webhook URLs
		Connect/disconnect third-

Integrations	/settings/integrations	party apps
API Keys	/settings/api-keys	Generate and revoke API keys
Webhooks	/settings/webhooks	Configure outbound event webhooks
Notification Templates	/settings/notifications	Edit email/SMS templates per event
Feature Flags	/settings/features	Enable/disable features per environment
Audit Log	/settings/audit-log	Searchable log of all system actions
Backup & Restore	/settings/backup	Schedule and trigger backups

9. API Layer Overview

9.1 Core API Standards

- **Base URL:** /api/v1/
- **Format:** JSON request/response
- **Auth Header:** Authorization: Bearer <JWT>
- **Pagination:** ?page=0&size=20&sort=createdAt,desc
- **Error Response Format:**

```
{
  "status": 400,
  "error": "VALIDATION_ERROR",
  "message": "Product name is required",
  "timestamp": "2024-01-01T10:00:00Z",
  "path": "/api/v1/products"
}
```

9.2 Key API Groups

Group	Base Path	Key Operations
Auth	/api/v1/auth	login, refresh, logout, mfa
Products	/api/v1/products	CRUD, variants, pricing, media
Categories	/api/v1/categories	CRUD, tree, reorder
Inventory	/api/v1/inventory	stock, adjustments, transfers
Orders	/api/v1/orders	CRUD, status transition, split
Payments	/api/v1/payments	capture, refund, reconcile
Shipments	/api/v1/shipments	create, track, update
Returns	/api/v1/returns	create RMA, approve, inspect
Customers	/api/v1/customers	CRUD, segments, communication
Suppliers	/api/v1/suppliers	CRUD, RFQ, PO, GRN
Reports	/api/v1/reports	query, export, schedule
Users	/api/v1/users	CRUD, roles, permissions
Settings	/api/v1/settings	store, tax, shipping, integrations
Notifications	/api/v1/notifications	templates, logs, send

10. Notifications & Alerts

10.1 Customer-Facing Notifications

Event	Channel
Order Confirmed	Email + SMS
Payment Failed	Email + SMS
Order Shipped (with tracking link)	Email + SMS + Push
Out for Delivery	SMS + Push

Order Delivered	Email + SMS
Return Approved	Email
Refund Processed	Email + SMS
Coupon Expiry Reminder	Email
Abandoned Cart Reminder	Email + SMS
Back In Stock Alert	Email + Push

10.2 Internal / Staff Notifications

Event	Who Gets Notified
New Order Placed	Order Manager
Order Flagged for Fraud	Order Manager, Admin
Stock Below Reorder Point	Inventory Manager
PO Pending Approval	Admin, Purchase Manager
Return Request Submitted	Customer Support, Warehouse
Chargeback Raised	Finance Staff
Shipment Delivery Exception	Shipping Coordinator
Failed Payment Reconciliation	Finance Staff
System Error / Integration Failure	Super Admin

11. Reports & Analytics

Key Metrics to Track

Metric	Module
Gross Merchandise Value (GMV)	Finance, Sales
Net Revenue (after returns/refunds)	Finance

Order Fulfillment Rate	Operations
Average Order Value (AOV)	Sales
Cart Abandonment Rate	Marketing
Customer Lifetime Value (LTV)	CRM
Inventory Turnover Ratio	Inventory
Return Rate by Product	Returns
On-Time Delivery Rate	Shipping
Supplier Fill Rate	Procurement
Days Sales Outstanding (DSO)	Finance
Gross Margin %	Finance

12. Technology Stack Summary

Frontend (Angular)

Concern	Technology
Framework	Angular 17+ (Standalone Components)
UI Library	Angular Material / PrimeNG
State Management	NgRx or Signals
HTTP Client	Angular HttpClient with interceptors
Routing	Angular Router with lazy loading
Auth Guard	CanActivate guards with role checks
Charts	ApexCharts / Chart.js / ECharts
Forms	Reactive Forms with validators
Internationalization	ngx-translate
Real-time	WebSocket (SockJS + StompJS)

Backend (Spring Boot)

Concern	Technology
Framework	Spring Boot 3.x
Security	Spring Security + JWT
ORM	Spring Data JPA + Hibernate
Database	PostgreSQL (primary), Redis (cache/session)
Search	Elasticsearch (product/order search)
Messaging	Apache Kafka / RabbitMQ (events)
File Storage	AWS S3 / MinIO
Email	Spring Mail + Sendgrid/SES
API Docs	SpringDoc / OpenAPI 3 (Swagger UI)
Validation	Hibernate Validator (Bean Validation)
Scheduling	Spring Scheduler / Quartz
Monitoring	Spring Actuator + Prometheus + Grafana

Infrastructure

Concern	Technology
Containerization	Docker + Docker Compose
Orchestration	Kubernetes (prod)
API Gateway	Kong / AWS API Gateway
CI/CD	GitHub Actions / Jenkins
Secret Management	HashiCorp Vault / AWS Secrets Manager
Logging	ELK Stack (Elasticsearch, Logstash, Kibana)

This document serves as the complete conceptual blueprint for an Enterprise E-Commerce Business Management Application built on Angular and Spring Boot. It covers all modules, their relationships, statuses, security requirements, roles, and UI structure — providing a solid foundation before development begins.